Abubakar Khawaja

Senior Full-Stack Engineer | React | Next.js | TypeScript | Python | Django | AWS

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PROFESSIONAL SUMMARY

Full-Stack Engineer with 5+ years of experience architecting and shipping production software across Python and Django, React, TypeScript, and PostgreSQL. I take features from first idea to live deployment, prioritizing clean UI architecture, highly concurrent APIs, and asynchronous pipelines that hold up under real-world load. I have run production services on AWS with Kubernetes and EKS, PostgreSQL on RDS, and Redis caching, and I have built agentic AI features with tool calling and vector retrieval. I am comfortable owning system design, database performance, and third-party integrations end to end.

EXPERIENCE

Senior Software Engineer | Codeupnow Sept 2025 – Present

- Worked on a microservices system in Python and Django with a React frontend, and cut downtime by 30%.
- Reworked Python REST API code and cut response times by 40%, which also brought cloud costs down.
- Wrote unit and API tests for Django REST endpoints with pytest and the Django test framework, using fixtures and factories for test data, and ran them in CI before every merge.
- Deployed and tuned services on AWS ECS/EKS and Lambda to support scaling, build performance, and deployment reliability.
- Optimized frontend performance with lazy loading, code splitting, and image optimization.
- Built reusable component libraries with Storybook, keeping the UI consistent across several projects.
- Built real-time data pipelines and background job processing for work that runs asynchronously.
- Mentored junior engineers through regular code reviews and drove Agile feature delivery in tight collaboration with UX/UI designers and backend teams.

Senior Full-Stack Engineer | Arbisoft July 2021 – Apr 2026

- Owned end-to-end third-party integrations across multiple production platforms, including Stripe for payments, SendGrid for transactional emails, and Looker for BI analytics.
- Redesigned React frontend architecture, reducing unnecessary re-renders by 30% and improving page load times by 20-30% via memoization, code-splitting, and optimized state management.
- Implemented SSR, SSG, and ISR using Next.js while improving Core Web Vitals, accessibility, SEO, and overall application responsiveness.
- Elevated automated test coverage from 10% to over 60% (Jest, Cypress, Vitest) and integrated quality gates (ESLint, Husky), reducing production bugs by 25%.
- Reworked GitLab CI/CD pipelines with parallel jobs and layer caching, reducing deployment times by 40%.
- Owned Sentry monitoring to cut production error volume: error triage with session replay, release and performance tracking, Slack and GitHub integration, and PII-safe payload encryption that keeps sensitive data out of captured events.
- Used AI coding tools such as Cursor, Claude Code, Codex, GitHub Copilot, and ChatGPT for both autonomous multi-step tasks and hands-on pair-programming, applying them to PR review, architecture discussions, and custom internal tooling, while comparing model outputs using structured prompts.

SELECTED PROJECTS

Telehealth Platform · *Next.js + Django + PostgreSQL + Kubernetes · Codeupnow*

- Architected and maintained a HIPAA-compliant telehealth platform on Django, with PostgreSQL on AWS RDS and Kubernetes for container orchestration.
- Built the frontend on the Next.js App Router with React Server Components, keeping data fetching on the server.
- Optimized patient-provider matching workflows with SQL query tuning and Redis caching, and worked with data engineering on ETL pipelines for real-time data processing.

FixAlert, Maintenance Platform · *React + Django*

- Implemented Celery-based background processing, Stripe payment integration, and reporting services.
- Optimized backend performance by eliminating N+1 queries and improving database access patterns.
- Designed asynchronous Celery workflows for QR generation, bulk PDF exports, and automated report generation.

Landit, Career Platform · *React + Next.js + TypeScript*

- Built a custom visualization layer for Looker in React after Looker's support team said it could not be done, hosting the built bundles and registering them through Looker config so they appeared in the standard visualization dropdown alongside the built-in charts.
- Executed zero-downtime frontend framework migrations (Material UI v4 to v5, React Router v4 to v5).
- Built a streaming retrieval chat feature backed by pgvector embeddings, rendering model output token-by-token over semantically retrieved content.

CORE SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3/SCSS
Frontend: React.js, Next.js (App Router, Pages Router), Vue.js, TypeScript, Redux Toolkit, TanStack Query, React Hook Form, Tailwind CSS, Material UI, SCSS
Backend: Python (Django, Django REST Framework, FastAPI, Flask), Node.js (NestJS, Express), REST API design, GraphQL, webhooks, JWT & OAuth2 authentication, Celery & RabbitMQ (async jobs), WebSockets, Scrapy
Databases: PostgreSQL, MySQL, MongoDB, Redis, Prisma, Drizzle ORM, Django ORM, Database Indexing, Query Optimization, Caching
Cloud & DevOps: AWS (ECS, EKS, Lambda, S3, RDS), Google Cloud (Firebase), Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Datadog, Sentry, Linux, Git
Testing: Jest, Cypress, Vitest, React Testing Library, Pytest, Django Test Framework, Testcontainers, ESLint, Prettier, Husky
Integrations: Stripe (payments), SendGrid & Resend (email), Firebase Auth, Google Maps, PostHog & Google Analytics, Google Looker (BI), Grafana, JWT/OAuth2
AI/ML: LLM Application Architecture, AI Agents, Tool/Function Calling, Structured Outputs (Zod, Pydantic), Model Context Protocol (MCP), RAG, Embeddings & Vector Databases (pgvector), Prompt & Context Engineering, OpenAI / OpenRouter / Ollama (local models), Vercel AI SDK
AI Development: ChatGPT, GitHub Copilot, Cursor, Claude Code, Codex, Prompt Engineering, AI-Assisted Development
Foundations: Data Structures & Algorithms, System Design, Performance Optimization, Agile/Scrum, Jira & Confluence, Code Review, Mentoring

EDUCATION

B.S. Software Engineering Punjab University College of Information & Technology (PUCIT) · GPA: 3.08

CERTIFICATIONS

- The AI Engineer Path & Prompt Engineering for Web Developers — Scrimba (July 2026)